Shio Runtime Architecture Specification

Version: 0.1 (Draft)

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Introduction

Shio Runtime is an architecture for building a believable digital character.

It is not centered around a specific AI model, rendering engine, speech synthesizer, desktop framework, or streaming platform.

Instead, the architecture is organized around clearly defined responsibilities.

Every subsystem exists for one purpose only.

Technologies may change.

Responsibilities must remain stable.

The purpose of this document is to define the architectural foundation of the project.

It describes the principles, responsibilities and interactions that every future implementation must follow.

This document is the primary source of truth for the architecture of Shio Runtime.

Whenever implementation and documentation disagree, the architecture defined in this document takes precedence until an intentional architectural decision is made.

Architectural Principles

The following principles define the architectural foundation of **Shio Runtime**.

Every architectural decision, every Work Item and every future subsystem must comply with these principles.

Principle 1 — Responsibilities Before Technologies

> **Shio Runtime is designed around responsibilities, not technologies.**

The architecture is organized around responsibilities rather than libraries, AI providers or frameworks.

LLM providers, TTS engines, ASR models, rendering engines and external services are implementation details.

Replacing them must never require redesigning the Runtime architecture.

Principle 2 — Runtime Executes, LLM Decides

> **LLMs make decisions. Runtime executes them.**

The language model never directly controls the application.

The LLM proposes intentions.

The Runtime validates them, executes them, manages permissions, coordinates state and guarantees system safety.

The Runtime always owns the final decision.

Principle 3 — One Runtime, One Responsibility

> **Every Runtime owns exactly one responsibility.**

Each Runtime exists to solve one clearly defined problem.

Responsibilities must never overlap.

If a Runtime starts accumulating unrelated logic, the architecture should be refactored before further development continues.

Principle 4 — Everything External Is an Adapter

> **Everything outside the Runtime is an adapter.**

External APIs, AI providers, rendering engines, databases, operating system integrations and future streaming platforms are adapters.

The Runtime defines contracts.

Adapters implement them.

Replacing an adapter must never require redesigning the Runtime.

Principle 5 — Experience Is As Important As Execution

> **Experience is as important as execution.**

Executing an action is only one part of creating a believable digital character.

Every meaningful action may produce an experience.

The Experience Layer decides how the character should react to completed actions.

Possible reactions include:

- speaking;
- remaining silent;
- changing facial expressions;
- performing gestures;
- triggering animations;
- intentionally showing no visible reaction.

Separating execution from experience allows Shio to behave like a coherent digital character instead of a collection of independent features.

Principle 6 — Preserve the Illusion

> **Every subsystem exists to preserve the illusion of a living character.**

Every Runtime contributes to the same goal.

Speech Runtime makes communication feel natural.

Avatar Runtime gives the character a believable body.

Experience Layer creates emotionally consistent reactions.

Memory Layer preserves continuity.

Action Runtime allows the character to interact with the outside world.

No subsystem exists for its own sake.

Every subsystem exists to strengthen the illusion that the user is interacting with a living digital character.

Principle 7 — The Runtime Exists to Disappear

> **The Runtime should never expose itself. Only the character should be visible.**

Users should never think about pipelines, controllers, adapters or AI providers.

They should not think:

> "The animation system started."

They should think:

> "Shio smiled."

They should not think:

> "The moderation tool banned the user."

They should think:

> "Shio kicked the toxic viewer."

The better the Runtime becomes, the less noticeable it is.

The Runtime exists to sustain the illusion of the character, not to expose its own implementation.

Character Philosophy

> **Shio is not an assistant controlling an avatar.**
>
> **Shio is a digital character whose body, voice, memory and actions are coordinated by the Shio Runtime.**

This distinction defines the entire architecture.

The Runtime exists to coordinate every subsystem required to create a believable digital character.

The character remains the center of every architectural decision.

Every future subsystem should strengthen the illusion of interacting with a living character rather than expose implementation details.

The Illusion Principle

The primary objective of Shio Runtime is not automation.

The primary objective is maintaining the illusion of life.

Every subsystem contributes to this illusion.

Breathing exists because living beings breathe.

Blinking exists because living beings blink.

Idle motion exists because living beings are never perfectly still.

Memory exists because living beings remember.

Experience exists because living beings react to the consequences of their own actions.

A feature that increases realism is often more valuable than a feature that increases complexity.

Whenever architectural decisions are made, preserving the illusion of a living character should be preferred over exposing technical implementation details.